Active Maintenance Report: coppock_guess_ternovski_2016

Coppock, Guess and Ternovski (2016, Political Behavior): When Treatments Are Tweets

2026-10-29

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This repository holds the actively maintained replication code for Coppock, Guess and Ternovski (2016), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

Article	10.1007/s11109-015-9308-6
Replication archive	10.7910/DVN/29548
Pre-analysis plan	None

The data are not redistributed here. The deposit is 797 KB across 6 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file against the checksum and byte size recorded in `original_manifest.csv`, and stops if `original/` holds anything the manifest does not list. The exact bytes this code was written against are therefore pinned in version control even though the bytes themselves are not. Checksums are quoted only from the manifest: a file this pipeline writes, such as a figure PDF, records the time it was written and hashes differently on every run.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `published_claims.csv` is the extraction: every numeric token in the article and its appendix, read off the pages and classified by hand. `extract_archive_values.R` runs the deposited archive in a scratch copy and records what it produces. `build_ground_truth.R` joins the two together with the rewrite's output. `maintained/in_text_claims.R` reaches the same numbers by a second, independent path and prints each one beside the sentence it comes from. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` corrects the published article, rendering to `coppock_guess_ternovski_2016_errata.pdf` at the root. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_guess_ternovski_2016.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 6 files, and produces every table and figure into `maintained/output/`. Required packages for the rewrite: tidyverse, estimatr, randomizr, nnet, modelsummary, knitr, kableExtra and here. Running the deposited archive needs AER, stargazer, xtable, lmtest and sandwich in addition, which `ground_truth/extract_archive_values.R` does in a scratch directory: set `ARCHIVE_RUN_DIR` to choose where, or leave it and the script uses a folder inside `tempdir()`. The deposit is never run in place, because running it writes files. Paths resolve through `here`, so nothing depends on the working directory. Most of the runtime is six randomization inference tests, each permuting assignment 1,000 times and refitting a pair of multinomial logits on every draw; every other script takes seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and a clean run leaves only the figure PDFs changed.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

No. Both analysis scripts stop with an error partway through. `LCV Experiments 1 and 2 analysis.R` dies at the Table 6 `stargazer()` call, taking Tables 7, 8, 9 and Figure 3 with it. `LCV Appendix analysis.R` dies earlier still, at the Study 1 omnibus balance test, because `complete_ra()` no longer accepts the argument `condition_names`; the current argument is `conditions`. Repair that and the appendix script dies at the Table A4 `stargazer()` call in turn, and one of its `print.xtable()` calls writes to a personal directory path that exists on no other machine.

Patch those three faults and everything computes, but one published number does not come back. The Study 1 omnibus balance p-value is 0.607 in the appendix and 0.579 here. It is a randomization inference p-value drawn with `set.seed(343)` over 1,000 permutations from `complete_ra()`, and the archive's own comments install randomizr from GitHub rather than from CRAN, pinned to nothing, so the random number stream that produced 0.607 belongs to a state of that repository a reader cannot now obtain. The Study 2 omnibus p-value has the same shape but a happier ending: 0.113 in the appendix, 0.117 under a current R, and 0.113 under `RNGkind(sample.kind = "Rounding")`. That one is the R 3.6 sampler change and nothing else, because Study 2's permutation function is written out by hand in the deposit instead of coming from a package.

1.2 Does the maintained rewrite reproduce the paper?

Yes, with 4 exceptions out of 660 published values, and none of them touches a result.

Two are the omnibus balance p-values, which are simulation output and are reported as such: 0.579 against a published 0.607 for Study 1, and 0.108 against a published 0.113 for Study 2, both far from any threshold and both within the Monte Carlo error of a 1,000-draw permutation test. The third is a count: the article says the 221 Study 2 petition signers "were followed by 1990 other users", and the deposited data hold 1991 subjects with a positive probability of exposure, of whom the 1975 whose probability is strictly below one are the sample Table 9 analyses. The fourth is the appendix's statement that subject types 5 through 7 make up "approximately 4.5%" of the Study 2 population, where the signing rate among direct message subjects is 4.41%.

Everything else matches to the precision the article prints: all of Tables 2 through 9, both balance tables, both heterogeneity tables, the account type shares in the Figure 3 caption, and every quantity the results sections state in running text as a number rather than as an inequality.

2 Paper overview

Citation: Coppock, A., Guess, A. and Ternovski, J. (2016). "When treatments are tweets: A network mobilization experiment over Twitter." *Political Behavior*, 38(1), 105-128. DOI: 10.1007/s11109-015-9308-6

Summary: Two field experiments run with the League of Conservation Voters test whether direct appeals over Twitter mobilize participation, and whether that mobilization spreads through the network. In each study LCV posted a public tweet asking supporters to sign an online petition, and subjects were randomly assigned to receive, in addition, a direct message addressing them as a "follower", a direct message addressing them as an "organizer", or nothing beyond the public tweet. Petition signers were then randomly shown a tweet button encouraging them to pass the link on, which randomly exposed the subjects' own followers to additional tweets and makes the second stage an experiment in its own right, analysed with inverse probability of exposure weights and instrumental variables. Study 1 (N = 6,687) ran in February 2014, Study 2 (N = 8,507) in July 2014. A direct message of either kind raised the signature rate by 3.6 percentage points in Study 1 and 4.4 in Study 2 while the public tweet moved nothing at all, and in Study 2 exposure to a signer's tweet raised signing among compliers by 5.6 percentage points.

3 Original archive reproducibility

Table 1: The deposited archive on R 4.6.1. LCVsource.r is not listed: it defines functions, is sourced by both scripts, and needs no changes.

Script	Top-level expressions	Error that stops source()	Expressions still failing when repaired and run one at a time
LCV Experiments 1 and 2 analysis.R	55	the condition has length > 1	2 (lines 142, 188)
LCV Appendix analysis.R	78	unused argument (condition_names = c("pub", "fol", "org"))	2 (lines 206, 241)

Every package the archive names still installs from CRAN: `stargazer`, `AER`, `ggplot2`, `dplyr`, `nnet`, `xtable` and `randomizr`, along with the `lmtest` and `sandwich` that `LCVsource.r` calls for by hand. The exposure is not availability but maintenance. The installed `stargazer` is version 5.2.3, dated 2022-03-03, and it carries the call-parsing defect described below. `randomizr` is under active development, and the argument the archive passes to `complete_ra()` is not one the installed version 2.0.1 accepts.

Running the two scripts expression by expression, so that one failure does not hide the next, separates the faults cleanly. Of the 8 `stargazer()` calls across the two files, 4 fail: the two exposure tables of the main text, which are Tables 6 and 9, and both of the appendix's heterogeneity tables, A4 and A5. The 4 that succeed deparse to a shorter call. Every other expression in the main script runs, including the whole Figure 3 block, which assembles all 32 plotted estimates and writes its PNG as intended. The appendix script adds two faults of its own, the `condition_names` argument and the hardcoded `print.xtable()` path. Run as shipped, the first of those also takes down the `hist()` and `abline()` calls that follow it, because the vector they plot was never filled; both edits together are what leaves only the two `stargazer()` failures in the table above.

3.1 The stargazer call-length failure

The error is `the condition has length > 1`, raised inside `stargazer`'s own `.get.object.names`. What decides it is neither the models nor the arguments:

```
short <- aaaaaaaaaaaaaaaaaaaaaaaa.1
stargazer(short, type = "text")                                          # fine
stargazer(aaaaaaaaaaaaaaaaaaaaaaaa.1, aaaaaaaaaaaaaaaaaaaaaaaa.1,
          type = "text")                                                 # fine
stargazer(aaaaaaaaaaaaaaaaaaaaaaaa.1, aaaaaaaaaaaaaaaaaaaaaaaa.1,
          aaaaaaaaaaaaaaaaaaaaaaaa.1, type = "text")                     # error
```

All three tabulate the same model, and the name is 26 characters long. `stargazer` deparses the call it was given in order to label the columns, and `deparse()` breaks its result into several strings once the text is long enough. The package then tests that vector with `if (is.na(s))`,

which was a warning before R 4.2 and has been an error since. Any archive that names its model objects descriptively and tabulates several of them at once is exposed, which is a large class of archives. The maintained rewrite tabulates with `modelsummary` and does not inherit the problem.

3.2 What still works

`prepareforgg()`, the Figure 3 helper, chains two `dplyr::do()` calls and takes the coefficients out of the second with `.$mod`. Both `do()` and that idiom are superseded, and the rewrite replaces them, but neither is broken: on dplyr 1.2.1 the chain runs and returns the correct eight rows per call. The reason Figure 3 never appears in a batch run is that the file stops at the third of its six `stargazer()` calls, well before the figure block and in a part of the script that has nothing to do with the figure.

3.3 The balance check that could not fail

The article reports an anomaly in Study 1: assignment to the tweet encouragement predicted petition signing at `p < .01`, which should have been impossible, since the encouragement was displayed only after a subject had signed. The deposit fits that regression with `lm()` and gets 0.0047; the rewrite fits it with HC2 errors and gets 0.0092. Both are below the threshold the article names, so the anomaly it reports is in the data rather than in the reporting of it.

The deposit runs the same check on Study 2 and its comment records no imbalance. That check cannot detect one. In Study 2 the encouragement was assigned inside the survey software after a subject clicked through, so `tweetbutton` is recorded only for the 221 petition signers, and within them the signing outcome is constant at one. The regression therefore has no outcome variation to explain, and it would return the same verdict whatever the data held. The design change is what rules out a Study 2 imbalance, not the regression.

4 Errata

One published label is wrong, and the deposit shows how. Table A3 reports a Study 2 covariate as “Number of Tweets” with arm means of 1559.503, 1554.428, 1552.586. The line that produces those numbers is `mean_days = mean(days_on_twitter, na.rm = TRUE)`, and the row label comes from a separate hand-written vector that says “Number of Tweets” where Table A2’s equivalent says “Days on Twitter”. The rewrite takes the mean of `days_on_twitter` within each arm and gets 1559.503, 1554.428, 1552.586, the same three numbers to every printed decimal. The row is days on Twitter, the values are right, and the label is a slip. The maintained rewrite labels it “Days on Twitter” and the ground truth records the values as matching.

Two entries of the reference list are wrong, and no ground truth row reaches them: the article gives the title of Bryan et al. (2011) as "Motivating voter turnout by invoking the sel", where the last word is "self", and the pages of Verplanken and Holland (2002) as "434", the first page of a range that runs to 447. Both were found by an audit that sends every printed reference entry whole to Crossref and checks the authoritative record back into it. Its first run flagged one other entry here, for a page range printed without its dash. That one is a false positive: the entry reads "pp. 12,653–12,656" on the page, with the dash present and the thousands separators the journal's style uses.

Table 8 also misspells "encouragement" in the label of its constant, where Table 5 prints the same label correctly. The values beneath it are as published.

All of these are set out in `coppock_guess_ternovski_2016_errata.pdf` at the root of this repository, as four numbered entries whose values are computed from the pipeline when the document is rendered. None changes a conclusion.

The archive's `strata_ra()` helper carries a second slip with no consequence: it writes `"ord"` where it means `"org"` when labelling the third arm of a simulated assignment. Since the labels only feed a multinomial logit whose deviance does not depend on them, the Study 2 omnibus p-value is unaffected. The rewrite permutes with `randomizr::block_ra()` instead, which draws the same block-by-block arm sizes.

5 Number-by-number comparison

What gets checked is fixed in advance, by an extraction rather than by what the pipeline happens to produce. `ground_truth/published_claims.csv` lists every numeric token in the article and its online appendix, read off the pages and classified by hand into 4 kinds: 705 are pipeline quantities, 20 are claims about a shape, a sign or a threshold rather than a value, and 13 are definitional or structural constants that no analysis can move. Every pipeline and descriptive claim must have both a row in the ground truth and a block in `maintained/in_text_claims.R`, and the build stops if either is missing.

The ground truth then pairs each published claim with two independently produced values: `value_script`, read out of the models the deposited code fits when it is run in a scratch copy of the archive, and `value_rewrite`, read out of `maintained/output/`. A published value enters only from the extraction, never from a script run, and the verdicts are computed by comparing the two against the string the article prints, at the precision it prints. A qualitative claim carries the same machinery with a truth value in place of a number, so an article contradicting its own estimates registers as a mismatch like any other. The complete table is `ground_truth/coppock_guess_ternovski_2016_ground_truth.csv`; the summary and every non-matching row are below.

Table 2: Ground truth coverage, by published float. Matches are counted against the rows that can be compared at all.

Published float	Rows	Archive matches	Rewrite matches
Table 1: organizations' Twitter accounts	4	0 / 0	0 / 0
Table 2: Study 1 design and outcomes	30	30 / 30	30 / 30
Table 3: Study 2 design and outcomes	35	35 / 35	35 / 35
Table 4: Study 1 direct message effects	60	60 / 60	58 / 58
Table 5: Study 1 tweet encouragement	16	16 / 16	16 / 16
Table 6: Study 1 network effects	30	30 / 30	30 / 30
Table 7: Study 2 direct message effects	60	60 / 60	58 / 58
Table 8: Study 2 tweet encouragement	16	16 / 16	16 / 16
Table 9: Study 2 network effects	30	30 / 30	30 / 30
Figure 1: Study 1 follower network	8	1 / 1	1 / 1
Figure 2: LCV tweets per weekday	2	0 / 0	0 / 0
Figure 3: effects by account type	13	12 / 12	12 / 12
Table A1: possible subject types	5	2 / 3	2 / 3
Table A2: Study 1 balance	51	49 / 51	49 / 50
Table A3: Study 2 balance	51	50 / 51	49 / 50
Table A4: Study 1 heterogeneity	136	136 / 136	116 / 116
Table A5: Study 2 heterogeneity	136	136 / 136	116 / 116
Figures A1 to A6: screenshots	1	0 / 0	0 / 0
Running text	57	38 / 39	38 / 39

Of the 741 recorded claims, 706 can be compared against a value the deposited code produces. 701 match and 5 do not. Every published float in the article and the appendix has a row here, including the ones nothing in the deposit can reach: the deposit is 2 data files, 3 code files and 1 README, with no stored results at all, so Table 1, Figure 1 and Figure 2 have no source in it. Those rows are listed as uncovered with the reason.

Table 3: Published claims recorded as not comparable, and why.

Float	Claim	Paper	Note
Running text	Study 1: p-value on the tweet encouragement assignment predicting signing		The article prints an inequality rather than a value, so there is nothing to compare at printed precision. The deposit's own lm() gives 0.0047 and the rewrite's HC2 fit gives 0.0092, both below the 0.01 the article asserts.
Table A1: possible subject types	Table A1: type 8 proportion, lower end of the printed interval	0.955	The appendix prints an interval derived from rounded inputs. Both computed complements, 0.9559 for Study 2 and 0.9637 for Study 1, fall inside it.
Table A1: possible subject types	Table A1: type 8 proportion, upper end of the printed interval	0.965	The appendix prints an interval derived from rounded inputs. Both computed complements, 0.9559 for Study 2 and 0.9637 for Study 1, fall inside it.
Table 1: organizations' Twitter accounts	Descriptive statistics of eleven environmental organizations' Twitter accounts		Collected from Twitter profiles in February 2015. The deposit holds only the two experimental data frames, so nothing in it can produce this table.
Table 1: organizations' Twitter accounts	Mean number of followers among the top ten organizations	118988	Derived from the printed table; no deposited data behind it.
Table 1: organizations' Twitter accounts	Mean tweets per weekday among the top ten organizations	13	Derived from the printed table; no deposited data behind it.
Table 1: organizations' Twitter accounts	Median account creation year among the top ten organizations	2008	Derived from the printed table; no deposited data behind it.

Table 3: Published claims recorded as not comparable, and why. *(continued)*

Float	Claim	Paper	Note
Figure 1: Study 1 follower network	Communities in the Study 1 follower network under the Walktrap algorithm	22	The network edge list is not deposited, so neither the figure nor its community count can be rebuilt.
Figure 1: Study 1 follower network	Modularity of the Study 1 follower network	0.3	The network edge list is not deposited.
Figure 1: Study 1 follower network	Possible connections between nodes in the Study 1 network	44709282	The network edge list is not deposited.
Figure 1: Study 1 follower network	Observed edges in the Study 1 network	131474	The network edge list is not deposited.
Figure 1: Study 1 follower network	Graph density of the Study 1 network	0.0029	The network edge list is not deposited.
Figure 1: Study 1 follower network	Median number of network members a follower follows	6	The network edge list is not deposited.
Figure 1: Study 1 follower network	Largest out-degree observed in the network	601	The network edge list is not deposited.
Figure 2: LCV tweets per weekday	LCV tweets and retweets per weekday, February 2013 to February 2015, mean	6.07	The daily tweet series is not deposited.
Figure 2: LCV tweets per weekday	LCV tweets and retweets per weekday, standard deviation	10.05	The daily tweet series is not deposited.
Running text	Intercoder reliability of the account type coding, Cohen's kappa	0.90	The coding was done by hand on 200 profiles and only the resolved account type is deposited, so the two coders' assignments no longer exist.
Running text	Study 1: users outside the LCV network who retweeted	5	Both data frames hold only LCV followers, so users outside the network are not in the deposit.
Running text	Study 2: users outside the LCV network who retweeted	7	Both data frames hold only LCV followers, so users outside the network are not in the deposit.
Running text	Total followers of the LCV's followers	7000000	Stated as an order of magnitude in the text. The follower-of-follower counts are not deposited.
Figures A1 to A6: screenshots	Screenshots of the petitions, the tweet encouragement and the public tweets		Six screenshots of experimental materials. No numbers and no data.
Figure 3: effects by account type	All 32 plotted conditional effects and their standard errors		The figure prints no numbers, so there is nothing to compare against the article. All 32 plotted estimate and standard error pairs agree with the estimates the deposit hands to ggplot to within 5.9e-15.

Where a table's `stargazer()` call fails, the models behind it still fit, and the archive values recorded here come from those model objects. The model objects are the informative comparison, because they separate a printing failure from an estimation failure, and here the failure is entirely in the printing.

A further 44 cells are recorded as unverifiable rather than as matches or mismatches. Each is a standard error on a term identified only within the public tweet arm, where no subject signed or tweeted: 0 of 3687 subjects in Study 1, and 0 of 3495 in Study 2. Every residual in that arm is therefore exactly zero, and the term's HC2 variance is a sum of zero squared residuals, which is zero up to floating point.

Whether the residue lands positive or negative is arbitrary, and it decides whether a standard error exists at all. `estimatr::lm_robust()` returns `NA` in 29 of these cells, where the residue is negative or small enough that estimatr treats the variance as zero, and a root of order 1e-11 in the other 15. A root that small prints as `0.000` and so would round to the published value, but reading that as agreement would make the verdict a report on a sign bit: 16 of these cells changed side between estimatr 1.0.6 and 2.0, 8 in each direction, on the same code and the

same data. Both signs are therefore recorded as unverifiable, which fixes the set at 44 cells instead of at whatever the arithmetic on a given machine happens to produce.

The deposit's verdicts are stable where the rewrite's are not, and the reason is a substitution rather than a better variance estimate. `vcovHC()` returns the same residue, and in 20 of these cells it comes back as `NaN`, which the deposit's own wrapper replaces with `0.00004`. Both `0.00004` and a tiny positive root print as `0.000`, so the archive reproduces its own printed table in all 44 cells whichever way the residue falls.

What the design supports for the public tweet arm is a bound rather than a point. Zero signatures from 3495 Study 2 subjects puts an exact 95 percent upper bound of 0.105 percent on that arm's signing rate, and zero from 3687 in Study 1 puts it at 0.100 percent. Neither figure is one the article prints, and neither is a replacement for the standard errors above, which are on regression terms rather than on a proportion.

6 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R` and 19 scripts covering one cleaning step, the eight published tables of the main text, the four appendix tables that come from data, Figure 3, two scripts for the quantities the article states only in running text, one that turns the article's qualitative claims into truth values, one that prints every published number beside the sentence it appears in, and one that re-runs the archive's own omnibus tests under the sampler R shipped before 3.6.0. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

6.1 Architecture

`clean_lcv.R` reads the two deposited data frames and writes tibbles with explicit factor levels for treatment and account type, and with the centred covariates flattened from the one-column matrices `scale()` leaves behind. Everything downstream reads those two files, so no analysis script touches `original/` directly.

The archive computes robust standard errors by fitting with `lm()` and then passing the fit through a `coeftest(fit, vcovHC(fit, type = "HC2"))` wrapper defined in `LCVsource.r`. The rewrite calls `estimatr::lm_robust()`, whose default `se_type` is already HC2, and `iv_robust()` in place of `AER::ivreg()` with the same wrapper. Those are the same estimators with the same standard errors.

Two places need care rather than translation. The article reports the follower against organizer difference by refitting on the two direct message arms alone; `lm()` silently drops the now-empty `pub` level and `lm_robust()` does not, so the rewrite calls `droplevels()` and the reference category comes out where the published table puts it. Figure 3 is the one output where the paper does not use robust standard errors: its intervals come from `coef(summary(lm(...)))`.

The figure script therefore passes `se_type = "classical"` explicitly, with a comment saying why, rather than quietly plotting a different interval from the one the article shows.

6.2 Deprecated patterns replaced

Table 4: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd("")‘	‘here::here()‘
‘lm()‘ + ‘coeftest(vcovHC, 'HC2')‘	‘estimatr::lm_robust()‘
‘AER::ivreg()‘ + the same wrapper	‘estimatr::iv_robust()‘
‘stargazer‘ (eight calls, four of them fatal)	‘modelsummary(output =)‘ plus a tidy ‘.csv‘
‘xtable‘ + ‘print.xtable‘ to a hardcoded path	‘knitr::kable()‘ to ‘output/‘
chained ‘do(mod = lm(...))‘ + ‘do(data.frame(...))‘	‘nest()‘ + ‘map()‘ + ‘unnest()‘
‘complete_ra(condition_names =)‘	‘complete_ra(conditions =)‘
hand-written ‘strata_ra()‘	‘randomizr::block_ra()‘
‘coord_flip()‘ + ‘geom_pointrange‘ + legend	estimate on ‘x‘, ‘geom_linerange‘, direct labels
magrittr pipe	native pipe

Nothing in `maintained/output/` is typed in. Every value there, including the two omnibus p-values and the four percentages in the Figure 3 caption, is computed from the deposited data by the script that writes it.

7 The omnibus balance tests

The appendix reports one omnibus balance p-value per study, each from a randomization inference procedure: fit a multinomial logit of assignment on all four covariates, take the likelihood ratio against an intercept-only fit, then repeat against 1,000 assignments permuted under the study's own protocol.

Table 5: Omnibus balance p-values, 1,000 permutations each. The old sampler column re-runs the deposit's own permutation code under RNGkind(sample.kind = "Rounding").

Study	Published	Archive	Archive, old sampler	Rewrite	Protocol
Study 1 (Table A2)	0.607	0.579	0.586	0.579	complete_ra(), from a 2015 GitHub randomizr
Study 2 (Table A3)	0.113	0.117	0.113	0.108	strata_ra(), written out in the deposit

Study 2's number is recoverable and Study 1's is not, and the difference is where the permutation comes from. Study 2 permutes with a function written out inside `LCVsource.r`, so the only thing standing between the published p-value and a current

R is the sampler: `RNGkind(sample.kind = "Rounding")` returns 0.113 exactly. Study 1 permutes with `complete_ra()`, which the archive's own comments say to install with `devtools::install_github("acoppock/randomizr")`. That names no version, and the released package has since dropped the very argument the archive passes it, so the draws behind 0.607 cannot be reconstructed, and the old sampler does not recover them either. The rewrite keeps the modern sampler in both cases and reports what it gets. Neither p-value is close to a threshold under any of the four routes above, so the balance conclusion the appendix draws is unaffected.

8 Figure 3

The figure plots conditional average treatment effects of the two direct message treatments within each account type, for each outcome and each study. The figure prints no numbers, so there is nothing to compare against the article. All 32 plotted estimate and standard error pairs agree with the estimates the deposit hands to ggplot to within 5.9e-15. What the article does print is the caption, which reports the account type composition of both samples, and all eight of those percentages reproduce.

Table 6: Figure 3 caption: sample composition by account type, in percent.

Quantity	Paper	Rewrite	Match
Study 1 sample male, percent	38.4	38.4	1
Study 1 sample female, percent	30.4	30.4	1
Study 1 sample organizations, percent	24.4	24.4	1
Study 1 sample unknown, percent	6.8	6.8	1
Study 2 sample male, percent	39.3	39.3	1
Study 2 sample female, percent	32.9	32.9	1
Study 2 sample organizations, percent	22.1	22.1	1
Study 2 sample unknown, percent	5.7	5.7	1

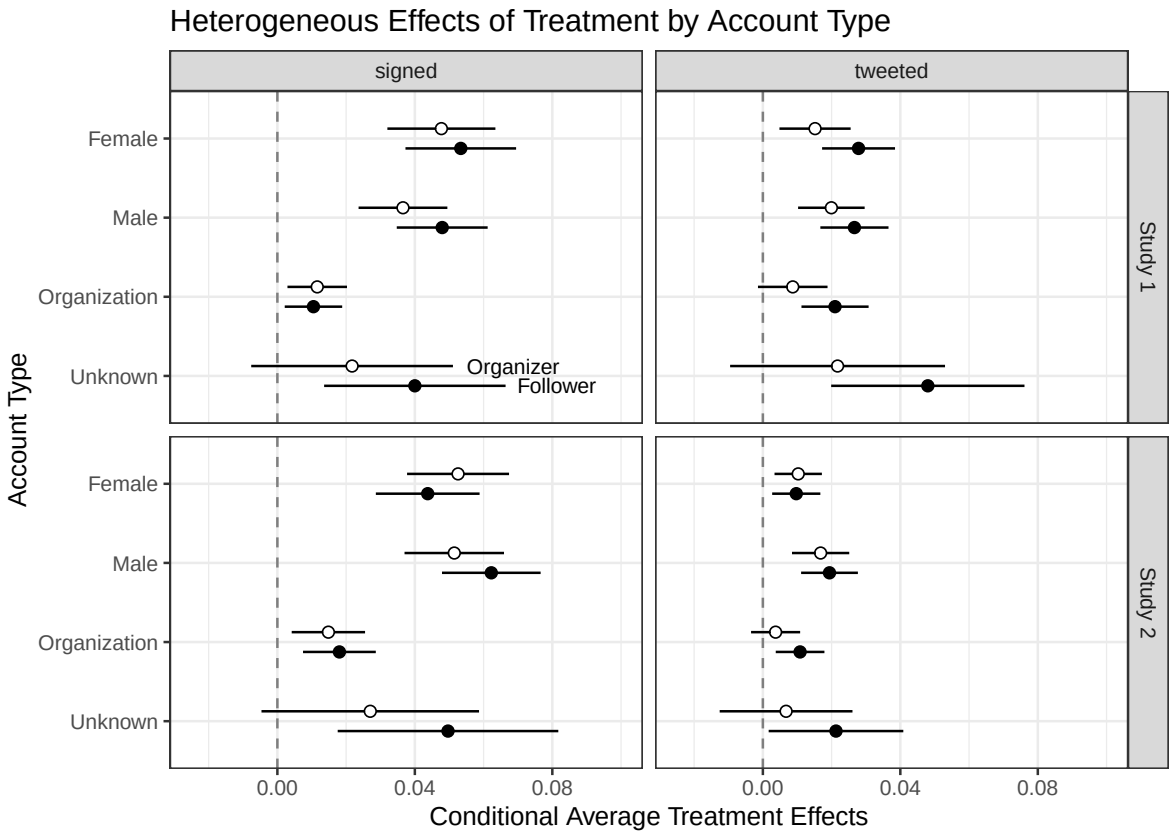


Figure 1: Figure 3 as reproduced by the maintained rewrite.

The rewrite's version differs from the published figure in presentation rather than content: the treatment arms are labelled beside the estimates instead of in a legend, the horizontal layout is built by mapping the account type to `y` rather than by flipping a vertical plot, and the interval is the fitted model's own 95 per cent confidence interval rather than the estimate plus or minus 1.96 standard errors.

9 Maintained rewrite verification

656 of 660 published values are reproduced to the precision the article prints. The 4 that are not are listed below with what separates them, and with where the fault lies.

Table 7: Published values the maintained rewrite does not reproduce.

Float	Claim	Paper	Rewrite	Fault	Note
Table A2: Study 1 balance	Omnibus p-value	0.607	0.579	archive	Randomization inference p-value over 1,000 permutations drawn with complete_ra(). The deposit's own comments install randomizr from GitHub rather than from CRAN and name no version, and the released package no longer accepts the argument the deposit passes it, so the stream behind the published value cannot be reconstructed, and the old sampler does not recover it either.
Table A3: Study 2 balance	Omnibus p-value	0.113	0.108	environment	Randomization inference p-value over 1,000 permutations. The deposit's own strata_ra() reproduces the published value exactly under RNGkind(sample.kind = 'Rounding'); both figures here use the sampler R has shipped since 3.6.0.
Running text	Study 2: subjects of the second-stage experiment	1990	1991	paper_internal	The deposited data hold 1991 Study 2 subjects with a positive probability of exposure, of whom the 1975 whose probability is strictly below one are the sample Table 9 analyses.
Table A1: possible subject types	Table A1: types 5 to 7 combined, Study 2, percent of population	4.5	4.409	paper_internal	The signing rate among Study 2 direct message subjects is 4.41 percent, which the appendix rounds up.

2 of the 4 sit in the article rather than in any code. The published follower count and the published subject type share are both stated in the text, and both disagree with the deposited data by a rounding or a transcription; the rewrite reports what the data hold. The Study 1 omnibus p-value is unrecoverable from the deposit for the reason given above, and the Study 2 one moves with the sampler that R shipped in 3.6.0, which is a property of the environment rather than a defect in either the archive or the rewrite.

10 In-text claims

Numbers stated in running prose are where a paper drifts from its own analysis, and a coverage check built around floats cannot see them. `maintained/in_text_claims.R` prints every published quantity beside the sentence it appears in, in the article's own units and to the article's own precision, section by section in reading order. It reads the same files in `maintained/output/` that the ground truth reads, and does its own filtering, unit conversion and rounding: two paths from one set of estimates to one claimed number, so a disagreement between them is a finding rather than a formality. It fits nothing, and 46 blocks between them cover every pipeline and descriptive claim in the extraction.

The 20 qualitative claims are handled separately, because they assert a shape rather than a value. Each is turned into a statement about the estimates and evaluated: that the organization conditional effects are the smallest of the four account types, that the follower prime is more than twice as effective as the organizer prime, that no subject in either public tweet arm signed or retweeted. 20 of 20 hold, and each carries the numbers behind the verdict in `maintained/output/text_descriptive_claims.csv`.

Table 8: The article's qualitative claims, evaluated against the estimates the maintained rewrite produces.

Claim	Holds	Evidence
d_encouragement_one_third	yes	Share shown the encouragement within each Study 1 direct message arm: 0.3333, 0.3333.
d_study1_pub_zero_signed	yes	Study 1 public tweet arm: 0 signatures among 3687 subjects.
d_study1_dm_significant	yes	Study 1 signing: follower 0.039 (p = 5.5e-15), organizer 0.033 (p = 7.2e-13); both positive and both below 0.01.
d_study1_encouragement_half	yes	Study 1 encouragement effect on tweeting among signers: 0.454, read as nearly half.
d_study1_organizer_38pp	yes	Encouragement by organizer interaction, Study 1: -0.384, that is 38.4 percentage points.
d_study1_follower_twice	yes	Study 1 encouragement effect: 62.4 percentage points in the follower arm against 24.0 in the organizer arm, a ratio of 2.60.
d_study1_network_null	yes	Study 1 second stage, columns 2 to 5, p-values: 0.46, 0.46, 0.73, 0.73.
d_study2_larger	yes	Signing effects: follower 0.039 in Study 1 against 0.046 in Study 2; organizer 0.033 against 0.043.
d_study2_follower_more_effective	yes	Study 2 tweeting: follower 0.014 against organizer 0.011, difference p = 0.3.
d_study2_tweeted_network_null	yes	Study 2 second stage, tweeting, columns 4 and 5, p-values: 0.18, 0.19.
d_study2_network_larger_than_main	yes	Study 2 complier effect on signing 0.056 against direct message effects of 0.046 and 0.043.
d_public_tweet_null	yes	Public tweet arms: Study 1 0 signed and 0 tweeted of 3687; Study 2 0 signed and 0 tweeted of 3495.
d_dm_click_4pp	yes	Effect of any direct message on signing: 3.6 percentage points in Study 1 and 4.4 in Study 2, both rounding to 4.
d_dm_signatures_exceed_tweets	yes	Study 1: signing 3.6 against tweeting 2.2 percentage points. Study 2: 4.4 against 1.3.
d_button_35_to_45	yes	Tweet encouragement effect on tweeting among signers: 35.0 percentage points in Study 2 and 45.4 in Study 1, a range of 35 to 45 at the precision the article prints.
d_study1_org_smaller_signed	yes	Study 1 signing: the organization conditional effect is the smallest of the four account types for both treatments, by -0.037 and -0.024 against the mean of the others.
d_study1_no_het_tweeted	yes	Study 1 tweeting, treatment by account type interactions, p-values: 0.91, 0.56, 0.53, 0.39, 0.33, 0.69.
d_study2_org_smaller_signed	yes	Study 2 signing: the organization conditional effect is the smallest of the four account types for both treatments, by -0.034 and -0.029. Study 2 tweeting interactions, p-values: 0.083, 0.23, 0.84, 0.13, 0.36, 0.64.
d_no_gender_pattern	yes	Male minus female conditional effects across the 8 study by outcome by treatment cells: 4 positive and 4 negative.
d_types_1_to_4_zero	yes	Signatures in the public tweet arms: 0 in Study 1 and 0 in Study 2.

11 R environment

Item	Value
R version	4.6.1
Platform	aarch64-apple-darwin23
Date run	2026-10-29

Table 10: Package versions used for the run behind this report.

Package	Version

estimatr	2.0.1.9000
randomizr	2.0.1
nnet	7.3.20
modelsummary	2.6.0
AER	1.2.17
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
here	1.0.2